A squared Lipschitz condition

Pugh, 3.1

Suppose $f : \mathbb{R} \to \mathbb{R}$ satisfies $|f(t) - f(x)| \leq |t - x|^2$ for all real t and x. Prove that f is constant.

Proof. We prove that $f(a) = f(b)$ for all $a, b \in \mathbb{R}$. The key step is to show, by induction on $n \in \mathbb{N}$, that for all $a, b \in \mathbb{R}$,

$$2^n |f(a) - f(b)| \leq |b - a|^2. \tag{$*$}$$

For $n = 0$, the left-hand side of $(*)$ is $|f(a) - f(b)|$. By hypothesis, with $t = a$ and $x = b$,

$$|f(a) - f(b)| \leq |a - b|^2 = |b - a|^2,$$

which proves $(*)$ for $n = 0$.

Now suppose $(*)$ holds for some n, for all $a, b \in \mathbb{R}$, and prove it for $n + 1$. Let $a, b \in \mathbb{R}$, and put

$$m = \frac{a + b}{2}.$$

Since

$$f(a) - f(b) = \big(f(a) - f(m)\big) + \big(f(m) - f(b)\big),$$

the triangle inequality gives

$$|f(a) - f(b)| \leq |f(a) - f(m)| + |f(m) - f(b)|.$$

Multiplying by $2^{n+1} \geq 0$, we obtain

$$\begin{aligned} 2^{n+1} |f(a) - f(b)| &\leq 2^{n+1} \big(|f(a) - f(m)| + |f(m) - f(b)|\big) \\ &= 2\big(2^n |f(a) - f(m)|\big) + 2\big(2^n |f(m) - f(b)|\big). \end{aligned}$$

Applying the induction hypothesis to the pairs a, m and m, b gives

$$2\big(2^n |f(a) - f(m)|\big) + 2\big(2^n |f(m) - f(b)|\big) \leq 2|m - a|^2 + 2|b - m|^2.$$

But

$$m - a = \frac{a + b}{2} - a = \frac{b - a}{2}, \qquad b - m = b - \frac{a + b}{2} = \frac{b - a}{2},$$

and $\left|\frac{b-a}{2}\right| = \frac{|b-a|}{2}$. Hence

$$2|m - a|^2 + 2|b - m|^2 = 2\left(\frac{|b - a|}{2}\right)^2 + 2\left(\frac{|b - a|}{2}\right)^2 = |b - a|^2.$$

Therefore

$$2^{n+1} |f(a) - f(b)| \leq |b - a|^2,$$

completing the induction. Thus $(*)$ holds for all $n \in \mathbb{N}$ and all $a, b \in \mathbb{R}$.

Now fix $a, b \in \mathbb{R}$. We show that $|f(a) - f(b)| = 0$. Since $|f(a) - f(b)| \geq 0$, it suffices to show that $|f(a) - f(b)| \leq 0$. Suppose instead that

$$|f(a) - f(b)| > 0.$$

By the Archimedean property, the powers of 2 are unbounded above, so there exists $n \in \mathbb{N}$ such that

$$2^n > \frac{|b-a|^2}{|f(a) - f(b)|}.$$

Since $|f(a) - f(b)| > 0$, this implies

$$|b-a|^2 < 2^n |f(a) - f(b)|,$$

contradicting $(*)$. Hence $|f(a) - f(b)| = 0$, so $f(a) - f(b) = 0$ and therefore $f(a) = f(b)$.

Since a and b were arbitrary, f is constant. $\square$